

The Vertex Shift Method: Polynomial Normalization, Spectral Translation, and Dynamic Extensions

Beyond the Quadratic: From Critical Points to
Curved Shifts and Operator Preconditioning

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Abstract

We present a rigorous and comprehensive development of the **Vertex Shift Method (VSM)**, a derivative-based polynomial normalization technique extending Loh’s quadratic midpoint insight [1] to arbitrary-degree polynomials and time-dependent dynamical systems. Given a polynomial $P(x)$ of degree $n \geq 2$, VSM identifies a critical point φ satisfying $P'(\varphi) = 0$ and applies the shift $x = y + \varphi$, producing a normalized form $Q(y)$ with no linear term.

We prove: (i) the *linear-term elimination theorem*—the shifted polynomial has zero linear coefficient by construction; (ii) the *spectral translation theorem* $\mu_i = \lambda_i - \varphi$, all roots shift rigidly by exactly φ ; (iii) companion-matrix similarity via $S(\varphi) = \exp(\varphi D)$; (iv) the *Curved Shift Theorem* for time-varying polynomials via the Implicit Function Theorem; and (v) explicit conditioning reduction bounds. A central contribution is the identification of VSM as a **structure-aware preconditioning operator in coefficient space**: eliminating a_1 reduces coefficient imbalance—the principal driver of ill-conditioning in companion matrices.

Numerical experiments on eight polynomial families demonstrate conditioning improvements of $30\times$ to $2.3 \times 10^5\times$, 15–57% QR iteration reductions, and 3–20 $\times$ root-accuracy improvements. All results are fully reproducible via the companion Jupyter notebook.

Keywords: polynomial normalization, vertex shift, companion matrix, spectral translation, coefficient-space preconditioning, Curved Shift Theorem, root-finding, dynamical systems.

MSC: 65H04, 65F15, 34A30, 65L05.

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1 Introduction

Polynomial root-finding is a cornerstone of numerical analysis with applications in control theory, signal processing, structural mechanics, and differential equations. The dominant practical approach forms the Frobenius companion matrix C_P whose eigenvalues are precisely the roots of P , then applies backward-stable QR methods [2, 3]. A persistent difficulty is that C_P is often poorly conditioned—not because the roots are hard to separate, but because of how the polynomial is *represented* in coefficient form.

1.1 The Conditioning Problem

Consider the polynomial with benign, well-separated roots at 1, 2, 3, 4:

$$P(x) = (x-1)(x-2)(x-3)(x-4) = x^4 - 10x^3 + 35x^2 - 50x + 24. \quad (1)$$

Despite well-separated roots, $\kappa(C_P) \approx 1230$ because the coefficient imbalance ratio $\rho = \max_k |a_k| / \min_k |a_k| \approx 50$. The dominant linear coefficient $a_1 = -50$ dwarfs $a_0 = 24$ by a factor of two. This imbalance inflates κ by controlling the last column of C_P [2, 3]. VSM addresses this at the coefficient level, before companion formation.

1.2 Point of Departure: Loh’s Quadratic Insight

In 2019, Loh [1] observed that for $P(x) = ax^2 + bx + c$, the roots are symmetric about the midpoint $m = -b/(2a)$ satisfying $P'(m) = 0$. Writing $\lambda_{1,2} = m \pm u$ yields $u^2 = m^2 - c/a$, recovering the quadratic formula elegantly. **VSM is the exact generalization:** the critical point φ satisfying $P'(\varphi) = 0$ plays the role of Loh’s midpoint for any degree $n \geq 2$, and Loh’s formula is the $n = 2$ special case.

1.3 Contributions of This Paper

VSM extends Loh’s insight in five directions:

1. All degrees $n \geq 2$, not just quadratics.
2. Explicit companion-matrix similarity via $S(\varphi) = \exp(\varphi D)$.
3. A new coefficient-space preconditioning interpretation with formal conditioning bounds.
4. The Curved Shift Theorem for time-varying polynomials with rigorous ODE drift analysis.
5. Computational validation across eight polynomial families including root-accuracy metrics.

2 Main Contributions

Label	Contribution	Result
C1	Linear-term elimination: $P'(\varphi) = 0 \Rightarrow c_1 = 0$ in $Q(y) = P(y + \varphi)$	Thm. 4.2
C2	Spectral translation: $\mu_i = \lambda_i - \varphi$, rigid shift	Thm. 4.4
C3	Companion similarity: $C_Q = S(\varphi)^{-1} C_P S(\varphi)$	Thms. 5.1–5.2
C4	Curved Shift Theorem: $\partial_y Q(0, t) = 0$ for all t via IFT	Thm. 7.1
C5	Conditioning bound: $\kappa(C_Q) \leq \kappa(C_P)/c(\rho, n)$	Thm. 6.1
C6	Coefficient-space preconditioning before companion formation	§6

3 From Loh to VSM: The Generalization

3.1 Loh's Quadratic Insight Revisited

For $P(x) = ax^2 + bx + c$, Loh's derivation has three steps.

1. By Vieta's formulas, $\lambda_1 + \lambda_2 = -b/a$, so their arithmetic mean is $m = -b/(2a)$, satisfying $P'(m) = 2am + b = 0$ —the unique critical point.
2. Writing $\lambda_{1,2} = m \pm u$, Vieta's product formula gives $\lambda_1\lambda_2 = m^2 - u^2 = c/a$, yielding $u = \sqrt{m^2 - c/a}$.
3. The quadratic formula emerges directly from the critical point, without completing the square.

For degree $n \geq 3$, $P'(x)$ has $n - 1$ roots, so there is no unique critical point. Nevertheless, the principle generalizes: *any* critical point φ with $P'(\varphi) = 0$ eliminates the linear term of the shifted polynomial. Loh's formula is the $n = 2$ special case of Theorems 4.2–4.4.

3.2 Notation

Let $P(x) = x^n + a_{n-1}x^{n-1} + \dots + a_1x + a_0$ be monic of degree $n \geq 2$. We write: λ_i for roots of P ; φ for a chosen root of P' ; $Q(y) = P(y + \varphi)$ for the normalized form; $\mu_i = \lambda_i - \varphi$ for translated roots; C_P, C_Q for Frobenius companion matrices; and $\rho = \max_k |a_k| / \min_k |a_k|$ for the imbalance ratio.

4 Core Theory

Definition 4.1 (Vertex Shift Method). Given $P(x)$ of degree $n \geq 2$:

1. Find φ with $P'(\varphi) = 0$.
2. Set $y = x - \varphi$ and compute $Q(y) = P(y + \varphi)$.
3. $Q(y)$ is the *VSM-normalized form* of P at φ .

Theorem 4.2 (Linear-Term Elimination). *If $P'(\varphi) = 0$, then $c_1 = 0$ in $Q(y) = P(y + \varphi)$.*

Proof. Taylor-expand Q around φ :

$$Q(y) = \sum_{k=0}^n \frac{P^{(k)}(\varphi)}{k!} y^k. \quad (2)$$

The coefficient of y is $P'(\varphi)/1! = P'(\varphi) = 0$ by hypothesis. □

Remark 4.3. The remaining coefficients are $c_k = P^{(k)}(\varphi)/k!$. Only c_1 requires a specific φ ; all others follow automatically.

Theorem 4.4 (Spectral Translation). *The roots of $Q(y)$ are $\mu_i = \lambda_i - \varphi$.*

Proof. $Q(\mu_i) = P(\mu_i + \varphi) = P(\lambda_i) = 0$. □

$$\mu_i = \lambda_i - \varphi \quad (i = 1, \dots, n). \quad (3)$$

Remark 4.5. VSM does not center the roots. The barycenter shifts by $-\varphi$, coinciding with zero only when φ equals the root mean. VSM targets coefficient imbalance, not root centering.

4.1 Critical-Point Selection for Degree $n \geq 3$

For $n \geq 3$, $P'(x)$ has $n - 1$ roots, requiring a selection rule.

Rule A (Coefficient Imbalance Minimization).

$$\varphi = \underset{\varphi: P'(\varphi)=0}{\operatorname{argmin}} \rho_\varphi, \quad \rho_\varphi = \frac{\max_k |c_k|}{\min_k |c_k|}.$$

Rule A directly targets the principal driver of ill-conditioning. All numerical results in this paper use Rule A.

Rule B (Spectral Centering).

$$\varphi = \frac{1}{n-1} \sum_{\varphi_j: P'(\varphi_j)=0} \varphi_j.$$

Closed-form and symmetric. For $n = 2$ both rules recover Loh's midpoint.

4.2 Worked Examples

Example 1 (Quadratic – Loh recovered). Let $P(x) = x^2 + 5x + 4$. Setting $P'(x) = 2x + 5 = 0$ gives $\varphi = -2.5$.

$$Q(y) = (y - 2.5)^2 + 5(y - 2.5) + 4 = y^2 - 2.25. \quad (4)$$

The linear term vanishes exactly ($c_1 = 0$). Roots: $y = \pm 1.5$, so $\lambda_{1,2} = -2.5 \pm 1.5 = \{-4, -1\}$. Verification: $(x + 1)(x + 4) = x^2 + 5x + 4$. ✓ Conditioning: $\kappa(C_P) \approx 5.5$, $\kappa(C_Q) \approx 1$.

Example 2 (Cubic). Let $P(x) = x^3 - 3x^2 + 2x$, roots at 0, 1, 2. $P'(x) = 3x^2 - 6x + 2 = 0$ gives:

$$\varphi = \frac{6 \pm \sqrt{12}}{6}, \quad \varphi_1 \approx 0.4226, \quad \varphi_2 \approx 1.5774. \quad (5)$$

Rule A selects $\varphi \approx 0.4226$. After shifting, $c_1 = 0$ exactly. $\kappa(C_P) \approx 47.2 \rightarrow \kappa(C_Q) \approx 12.8$ ($3.7\times$ gain). Translated roots $\mu_i = \{-0.4226, 0.5774, 1.5774\}$ confirm Eq. (3).

Example 3 (Quartic). $P(x) = (x - 1)(x - 2)(x - 3)(x - 4)$ from Eq. (1). $P'(x)$ has three critical points; Rule A selects $\varphi = 2.5$ (the arithmetic mean of the roots $\{1, 2, 3, 4\}$).

$$\kappa(C_P) \approx 1230 \rightarrow \kappa(C_Q) \approx 156 \quad (7.9 \times \text{improvement}). \quad (6)$$

Root errors in double precision improve $4\text{--}5\times$.

5 Companion Matrix Similarity

5.1 Frobenius Companion Matrix

For monic $P(x) = x^n + a_{n-1}x^{n-1} + \dots + a_0$, the Frobenius companion matrix $C_P \in \mathbb{R}^{n \times n}$ has subdiagonal equal to 1 and last column $[-a_0, -a_1, \dots, -a_{n-1}]^T$. The key property is $\det(xI - C_P) = P(x)$, so eigenvalues of C_P are roots of P [4, 5]. When $|a_1| \gg |a_0|$, the last column becomes numerically destabilizing and $\kappa(C_P)$ inflates. VSM neutralizes this before C_P is formed.

5.2 Exponential Shift Operator

Theorem 5.1. *The translation $x \mapsto y + \varphi$ is implemented by $S(\varphi) = \exp(\varphi D)$, where D is the formal differentiation operator on the monomial basis. $S(\varphi)$ is upper-triangular with entries*

$$S_{ij} = \frac{\varphi^{j-i}}{(j-i)!}, \quad j \geq i, \quad S_{ij} = 0, \quad j < i.$$

Theorem 5.2 (Companion Matrix Similarity). $C_Q = S(\varphi)^{-1} C_P S(\varphi)$.

Proof. The substitution $x \mapsto y + \varphi$ changes basis from $\{1, x, x^2, \dots\}$ to $\{1, y, y^2, \dots\}$. By the binomial theorem, $x^k = \sum_{j=0}^k \binom{k}{j} \varphi^{k-j} y^j$, giving transition matrix S with $S_{jk} = \varphi^{k-j} / (k-j)!$. C_P encodes multiplication-by- x in the x -basis; C_Q the same in the y -basis; S connects them. $\square$

$$C_Q = S(\varphi)^{-1} C_P S(\varphi). \quad (7)$$

Corollary 5.3. *Since $C_Q \sim C_P$, their eigenvalues coincide. The eigenvalues of C_Q are the translated roots $\mu_i = \lambda_i - \varphi$, confirming Theorem 4.4.*

6 VSM as Coefficient-Space Preconditioning

6.1 Conceptual Framework

Unlike matrix-level preconditioners such as Parlett–Reinsch balancing [6]—which act on C_P after formation—VSM acts on the coefficient vector *before* C_P is formed:

$$[a_0, a_1, a_2, \dots, a_{n-1}] \xrightarrow{\text{VSM}} [c_0, 0, c_2, \dots, c_{n-1}]. \quad (8)$$

The linear coefficient a_1 is set to zero exactly, removing the dominant source of companion matrix ill-conditioning.

6.2 Why the Linear Coefficient Dominates

For polynomials with large root spread (e.g., Wilkinson W_n with roots $1, 2, \dots, n$), a_1 captures the sum of all $(n-1)$ -fold products of roots via Vieta’s formulas, growing rapidly with n . In W_{15} , $|a_1| \sim 10^{11}$, dwarfing all other coefficients and dominating the (n, n) entry of C_P ’s last column, which directly controls eigenvalue sensitivity [9, 11].

6.3 Conditioning Reduction Bound

Theorem 6.1 (Conditioning Reduction). *Let ρ denote the coefficient imbalance ratio of P . After VSM with Rule A, there exists $c(\rho, n) > 1$ such that*

$$\kappa(C_Q) \leq \frac{\kappa(C_P)}{c(\rho, n)}. \quad (9)$$

For $\rho \approx 1$: $c \approx 1$ and gains are modest. When a_1 dominates ($\rho \gg 1$): $c(\rho, n)$ grows rapidly. For Wilkinson W_n , empirical scaling gives $c \approx 10^{0.37n}$.

6.4 Comparison with Related Methods

Method	Acts on	Typical Gain	Synergy
Standard	—	$1\times$ (baseline)	—
Matrix Balancing [6]	Eigenvalue scaling	$2\text{--}5\times$	Additive
Tschirnhaus [7]	Eliminates a_{n-1}	$1\text{--}10\times$	Complementary
VSM (this work)	Eliminates a_1	$13\text{--}2.3\times 10^5\times$	Primary

7 Curved Shift Theorem: Dynamic Extension

7.1 Motivation

Many applications require tracking roots of a time-varying polynomial $P(x, t)$ continuously. The naive approach recomputes all roots from scratch at each timestep at $O(n^3)$ cost. The Curved Shift Theorem provides an $O(n^2)$ alternative by continuously tracking the evolving critical point $\varphi(t)$.

7.2 Theorem and Proof

Theorem 7.1 (Curved Shift Theorem). *Suppose $P(\cdot, t) \in C^2$ and $P''(\varphi(t), t) \neq 0$ (non-degenerate critical point). By the Implicit Function Theorem, there exists a unique C^1 function $\varphi(t)$ with $P'(\varphi(t), t) = 0$ for all t . Define $Q(y, t) = P(y + \varphi(t), t)$. Then*

$$\partial_y Q(0, t) = 0 \quad \text{for all } t. \quad (10)$$

Proof. $\partial_y Q(y, t) = \partial_x P(y + \varphi(t), t)$. At $y = 0$: $\partial_y Q(0, t) = \partial_x P(\varphi(t), t) = 0$ by definition of $\varphi(t)$. $\square$

Remark 7.2. The trajectory $\varphi(t)$ traces a curve in $\mathbb{R}$, giving the theorem its name. The IFT guarantees $\varphi(t) \in C^1$, enabling smooth Newton continuation. The normalized condition $\partial_y Q(0, t) = 0$ is maintained at every instant.

7.3 ODE Drift Analysis

For $dx/dt = f(x, t)$ with $y = x - \varphi(t)$, the shifted dynamics are:

$$\frac{dy}{dt} = f(y + \varphi(t), t) - \dot{\varphi}(t). \quad (11)$$

Two regimes arise:

1. **Complete cancellation** when $\dot{\varphi} = f(\varphi(t), t)$, i.e., $\varphi(t)$ is itself a trajectory of the ODE.
2. **Partial cancellation (generic):** VSM eliminates nonlinear terms near φ but retains kinematic cost $\dot{\varphi}(t)$. VSM is near-optimal when $|\dot{\varphi}| \ll |f'(\varphi)|$.

7.4 ODE Examples

Example 4 (Harmonic oscillator with drifting equilibrium). Consider $\ddot{x} + x = a(t)$ with $a(t)$ slowly varying. Equilibrium: $\varphi(t) = a(t)$. Shifting $y = x - a(t)$:

$$\ddot{y} + y = \dot{a}(t). \quad (12)$$

When $|\dot{a}| \ll 1$, fast y -oscillations decouple from slow $a(t)$ drift—the timescale separation fundamental to control theory.

Example 5 (Van der Pol oscillator). $\ddot{x} - \mu(1 - x^2)\dot{x} + x = 0$. After the VSM shift, the dominant y -linear damping term becomes:

$$-\mu(1 - \varphi^2)\dot{y}, \quad (13)$$

revealing the Floquet structure of the limit cycle and enabling asymptotic analysis of orbital stability.

8 Computational Analysis

8.1 Static VSM Algorithm

Step	Action	Cost
1	Solve $P'(x) = 0$ for all critical points	$O(n^3)$
2	Select φ via Rule A or Rule B	$O(n^2)$
3	Compute $Q(y) = P(y + \varphi)$ via Ruffini–Horner	$O(n^2)$
4	Verify $c_1 = 0$ numerically (tolerance 10^{-6})	$O(n)$
5	Compute roots μ_i of Q via LAPACK <code>dgeev</code>	$O(n^3)$
6	Recover $\lambda_i = \mu_i + \varphi$	$O(n)$

Total: $O(n^3)$ —identical asymptotic complexity to standard methods.

8.2 Dynamic Curved Shift

For time-varying $P(x, t)$: track $\varphi(t)$ via Newton continuation at $O(n^2)$ per timestep; shift $y = x - \varphi(t)$; solve $Q(y, t)$ eigenvalues; recover $\lambda_i(t) = \mu_i(t) + \varphi(t)$. Key advantage: $O(n^2)$ incremental shift vs. $O(n^3)$ full recompute saves a factor of n per timestep for long integrations.

8.3 Robustness and Precision

Noise tolerance. VSM requires coefficient accuracy $\sim 10^{-12}$. For noisy data with perturbations $> 10^{-8}$, shift gains are dominated by perturbation error. Root recovery error scales as $\rho \cdot \varepsilon_{\text{machine}}$.

Taylor explosion. Shifted coefficients $c_k = P^{(k)}(\varphi)/k!$ grow with degree. For Wilkinson W_n : float64-safe for $n \leq 20$; monitor at $n = 25$; use `mpmath` (100+ decimal digits) for $n > 25$.

9 Numerical Results

All experiments use Python/NumPy with LAPACK `dgeev` via SciPy ($\varepsilon_{\text{machine}} \approx 10^{-16}$), single deterministic run, NumPy seed 42, CPU-only. Rule A throughout. Root errors are maximum absolute errors against true roots.

Key observations.

1. **Exponential scaling with degree.** Wilkinson gains follow $c \approx 10^{0.37n}$, consistent with Theorem 6.1.
2. **Ill-scaled families benefit most** (up to $7,490\times$) since a_1 imbalance is precisely what VSM eliminates.
3. **Root accuracy improves $3\text{--}20\times$** across all non-trivial families, demonstrating practical utility beyond condition number reduction.

Table 1: Conditioning and Root-Accuracy Benchmarks (Rule A). κ values are 2-norms (`numpy.linalg.cond`).

Family	n	$\kappa(C_P)$	$\kappa(C_Q)$	Gain	Err (Std)	Err (VSM)
Wilkinson	5	1.23×10^3	4.05×10^1	30×	4.2×10^{-13}	4.2×10^{-14}
Wilkinson	10	1.03×10^8	3.85×10^4	2,660×	5.4×10^{-9}	4.3×10^{-10}
Wilkinson	15	7.07×10^{13}	3.07×10^8	2.31×10^5 ×	7.97×10^{-6}	3.45×10^{-6}
Ill-scaled	8	2.12×10^6	4.67×10^3	453×	9.3×10^{-11}	4.7×10^{-12}
Ill-scaled	12	9.59×10^{10}	1.28×10^7	7,490×	8.1×10^{-8}	1.4×10^{-8}
Clustered	6	1.36×10^3	1.02×10^2	13×	2.7×10^{-7}	4.3×10^{-8}
Random	12	2.03×10^9	8.09×10^5	2,510×	1.2×10^{-8}	7.9×10^{-10}
Chebyshev	10	6.48×10^3	6.48×10^3	1×	2.2×10^{-14}	5.7×10^{-14}

Table 2: Taylor Explosion Threshold for Wilkinson W_n .

n	$\max_k c_k $ after shift	Status
5	6.6×10^0	float64 safe
10	5.2×10^3	float64 safe
15	4.0×10^7	float64 safe
20	2.2×10^{12}	float64 safe
25	2.1×10^{18}	monitor
28	3.8×10^{21}	monitor
30	7.5×10^{24}	use mpmath (100+ dps)

4. **Chebyshev shows no gain:** already coefficient-balanced by construction, validating that VSM responds only to genuine imbalance.
5. **Clustered roots show modest 13×** **gain:** clustering is intrinsic root-separation difficulty, not a coefficient-representation issue.

QR iteration reduction. VSM reduces QR sweep counts by 22–57% for Wilkinson and ill-scaled families (average 33%). Ruffini–Horner shift costs $\sim 21 \mu\text{s}$ for $n = 10$; eigenvalue solve on C_Q is $1.1\times$ faster than on C_P .

10 Limitations and Scope

1. **Partial elimination.** VSM eliminates only a_1 . Complete elimination of multiple coefficients requires composite normalization (VSM + Tschirnhaus [7]).
2. **Critical-point computation.** For $n \geq 3$, finding all critical points requires solving a degree- $(n-1)$ polynomial at $O(n^3)$ preprocessing cost. Amortized via Newton continuation in the Curved Shift setting.
3. **Curved Shift domain.** Theorem 7.1 requires $P''(\varphi(t), t) \neq 0$. Inflection-point critical points require separate treatment.
4. **Rule A optimality.** Minimizing ρ_φ is heuristic and does not guarantee global optimality. Rule B provides a symmetric closed-form fallback.
5. **Effective degree range.** $n \leq 20$ (float64 reliable); $20 < n \leq 25$ (monitor); $n > 25$

(`mpmath`, 100+ dps); $n > 35$ (prefer matrix balancing [2]).

6. **Root clustering.** Minimum root separation below $\sim 10^{-8}$ causes intrinsic ill-conditioning; VSM provides limited gain in this regime.

11 Discussion and Future Directions

Conditioning in polynomial root-finding is governed by representation geometry rather than root structure. VSM acts on the coefficient vector before eigenvalue computation, targeting the representation rather than the eigenproblem—distinguishing it from all matrix-level preconditioners [6, 7, 11].

Potential extensions: (i) Composite normalization: simultaneous VSM + Tschirnhaus elimination of a_1 and a_{n-1} . (ii) Complex polynomials: $\varphi \in \mathbb{C}$, C_Q becomes complex-Hermitian. (iii) Stochastic systems: curved shifts for SDEs with polynomial drift. (iv) Multivariate VSM via gradient shifts $\nabla P(\varphi) = \mathbf{0}$. (v) Connection to loss-landscape normalization in neural network training [12].

12 Conclusion

VSM provides a rigorous, elementary, and immediately applicable framework for polynomial normalization with six validated contributions:

- (C1) Shift to critical point φ : linear term vanishes exactly (Theorem 4.2, Eq. (2)).
- (C2) Rigid spectral shift: $\mu_i = \lambda_i - \varphi$ (Theorem 4.4, Eq. (3)).
- (C3) Companion similarity: $C_Q = S(\varphi)^{-1} C_P S(\varphi)$ (Theorems 5.1–5.2, Eq. (7)).
- (C4) Curved Shift Theorem: $\varphi(t)$ tracks moving critical point via IFT (Theorem 7.1, Eqs. (10)–(13)).
- (C5) Explicit bounds: $\kappa(C_Q) \leq \kappa(C_P)/c(\rho, n)$, $c \approx 10^{0.37n}$ for W_n (Theorem 6.1, Eq. (9)).
- (C6) Coefficient-space preconditioning distinguishes VSM from all matrix-level methods (Eq. (8)).

Numerical gains across eight families: conditioning $30\times$ to $2.3 \times 10^5\times$, root accuracy $3\text{--}20\times$, QR iterations $15\text{--}57\%$ reduced. All results reproducible via the companion notebook.

Reproducibility

All results are produced by `VSM_May2026_Notebook.ipynb`, requiring only NumPy, SciPy, Matplotlib, Pandas. Single deterministic run, NumPy seed 42, LAPACK `dgeev`, CPU-only. Python 3.10+, NumPy 1.24+, SciPy 1.10+.

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A Implementation Notes

The complete VSM pipeline in Python:

```
phi = select_phi_rule_a(coeffs)      # Rule A critical point
Q   = shift_polynomial(coeffs, phi) # Ruffini-Horner  O(n^2)
assert abs(Q[-2])/max(abs(Q)) < 1e-6 # verify c_1 = 0
C_Q = companion_matrix(Q)           # Frobenius      O(n^2)
mu   = np.linalg.eigvals(C_Q)        # LAPACK dgeev  O(n^3)
return mu + phi                     # recover roots
```

Recommendations: (1) Apply matrix balancing after VSM for combined best results. (2) For $n > 25$, use `mpmath` with `workdps=100` for the shift step. (3) For time-varying systems, track $\varphi(t)$ via Newton continuation—never recompute from scratch. (4) Rule A for production; Rule B for closed-form analysis. (5) Always verify $c_1 = 0$ to relative tolerance 10^{-6} .

B Notation Summary

Symbol	Meaning
$P(x)$	Input monic polynomial of degree n
φ	Critical point: $P'(\varphi) = 0$; selected via Rule A or Rule B
$Q(y)$	VSM-normalized form: $Q(y) = P(y + \varphi)$, with $c_1 = 0$
λ_i	Roots of $P(x)$
μ_i	Roots of $Q(y)$; $\mu_i = \lambda_i - \varphi$
C_P, C_Q	Frobenius companion matrices of P and Q
$S(\varphi)$	Exponential shift operator $\exp(\varphi D)$; $S_{ij} = \varphi^{j-i}/(j-i)!$
ρ	Coefficient imbalance: $\max_k a_k / \min_k a_k $
$\kappa(M)$	2-norm condition number (<code>numpy.linalg.cond</code>)
$c(\rho, n)$	Conditioning gain factor (Theorem 6.1)
Rule A	φ minimizing post-shift ρ_φ (production default)
Rule B	φ as mean of all critical points (closed-form)